

ADITYA VERMA

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EDUCATION

Year	Degree	Institute	CGPA
2021 - 2025	B.Tech in Computer Science and Engineering	Graphic Era University, Dehradun	8.32

EXPERIENCE

Software Engineer, Tata Consultancy Services - Mumbai, Maharashtra November, 2025- Current

- Architected Python-based backend systems for a large-scale system modernization initiative at the **Reserve Bank of India**, migrating **200+ workflows across 5,000+ institutions** from a legacy system to a distributed architecture.
- Built bi-directional data processing frameworks for structured data ingestion and dynamic report generation, powering **real-time query and enterprise-scale reporting systems**.
- Developed high-performance distributed data pipelines using **Python and PySpark**, writing SQL-based validation and transformation logic to ensure data reliability and consistency at scale.
- Led debugging and defect resolution across legacy-to-new system migrations, collaborating cross-functionally in an **Agile** environment to ensure consistent, production-grade behavior.

PROJECTS

WordSphere: A comprehensive blogging application (August, 2024 - September, 2024) : word-sphere.vercel.app

- Developed a full-stack blogging application using React.js for the frontend and Appwrite as a backend-as-a-service, configuring authentication, database collections, and storage buckets to handle user management and content persistence.
- Implemented full CRUD functionality for blog posts (creation, editing, deletion) along with advanced content filtering, leveraging Appwrite's REST APIs for secure, scalable data operations while deploying the application on Vercel, managing the project end-to-end from architecture and backend configuration to production deployment.

Self-Driving Car Agent (September, 2023 - December, 2023) : github.com/addy-301/Self-Driving-Car-Agent-with-RL

- Built an autonomous driving agent using Unity ML-Agents, training and benchmarking PPO and SAC reinforcement learning algorithms to **92% decision-making accuracy** and **3.5s average adaptation time** to dynamic conditions.
- Designed a custom simulation track with real-time sensor and actuator integration, validating end-to-end model performance via TensorBoard benchmarking.

Real-Time Vehicle Tracking System (August, 2023 - September, 2023) : github.com/addy-301/RealTime-Vehicle-Tracking-System-SIH2023

- Engineered a full-stack MERN application for real-time bus tracking, achieving **98% real-time location update accuracy** via Google Maps API integration; deployed on Azure and GitHub Pages.
- Implemented role-based authentication, live seat-availability tracking, and a one-tap SOS feature broadcasting live GPS coordinates to a control room.

SKILLS

Programming Languages	Python, Java, TypeScript, HTML, CSS, C++
Web Development	React.js, Node.js, Express.js, REST APIs, Tailwind CSS
Database & Backend	MySQL, MongoDB, Mongoose, Appwrite
AI / Machine Learning	Generative AI, Retrieval-Augmented Generation (RAG), Agentic AI, Reinforcement Learning,
Cloud & Tools	Microsoft Azure, Google Cloud Platform (GCP), Git, GitHub, Docker, Vercel
Core Concepts	SDLC, Agile, Object-Oriented Programming (OOP), Data Structures and Algorithms

ACHIEVEMENTS & CERTIFICATIONS

- TCS AI Hackathon** - Qualified Round 1 and Round 2 for an internal TCS hackathon, competing against 60 teams, building AI solutions for complex business problems in a limited time constraint placing our team in Regional finals for Mumbai region.
- Intra-University Coding Competition** - Secured 3rd rank in a coding contest hosted by Graphic Era University.
- TechWizard** - Qualified for Round 2 (twice) for an inter-university coding contest hosted by Graphic Era University.
- INFathon** - Qualified for Round 2 for a coding contest organized by Informatica.

Certifications: Salesforce Agentblazer Innovator (Salesforce) · AWS GenAI Practitioner (AWS) · Introduction to Generative AI (Google Cloud)
· Google Cloud Computing Fundamentals (Google Cloud)